

SILTHAUL · REQUIREMENTS

SiltHaul requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 status from SLH-CAL-001 on the constructable design (SLH-DDR-002); R10 restated against the value-engineering target
0.2	2026-10-03	Amish Chadha	TRL 2 requirements with concept status; R11 overload limit and R12 load holding added (SLH-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

SiltHaul requirements

Nine of the twelve requirements are met on paper or by design, one is at risk, one can only be verified by a timed trial and one is not met (SLH-CAL-001, Table 3). The miss is R1, output: about 0.32 m³ an hour against about 0.90 m³ an hour for a bucket crew of the same size, because two people at a crank can only give about 50 W each. SiltHaul's gain is that nobody lifts or carries mud out of the building. R11 and R12 were added at TRL 2 under Amish's pre-approval of 2026-10-03 (SLH-DDR-001), because the anchors and rope can only be sized on a known maximum rope tension, and a hand-cranked load must hold when the handles are let go. "Met on paper" means shown by calculation, not by test.

Safety: SiltHaul is a rope-and-anchor system under tension with a hand-cranked chain drive. R6, R7, R11 and R12 are its safety requirements; they are checked on paper only and must be proved by proof-load tests before anyone uses the kit (SLH-BLD-001, section 6).

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (SLH-CAL-001)
R1	Mud moved per hour by a crew of four	At least twice the volume moved by the same crew with shovels and buckets (estimate)	Side-by-side timed trial on a test bed of wet silt	Not met on paper: 0.32 against 0.90 m ³ /h (ratio 0.36); nobody carries mud
R2	Crank handle force at full box	150 N (34 lbf) or less per person with two people at the cranks	Spring balance at the handle during trial	Met on paper: 63 N each at 1,000 N (126 N for one person)
R3	Scraper box capacity	About 40 L (10 US gal) of wet mud per trip	Measured fill on test bed	Met by design: 40.7 L at a 165 mm fill
R4	Fits through a doorway	Box, tail block and ramp pass a 0.7 m (28 in) opening, with the return leg beside the box	Fit test	Met by design: 494 mm over the shackles; ramp 646 mm
R5	Working length	Up to 15 m (49 ft) from tail sheave to capstan	Trial at full length	Met by design: 12 m of travel on each drum half; capstan at least 6 m from the door
R6	Rope and anchor strength	Minimum break strength at least 5 times the working pull, and at least 5 times the R11 limit for the rope and chain	CalRig proof-load test of each anchor type	Met on paper: rope 18 times working, 5.4 at the limit; chain 5.0
R7	Tail anchor independent of walls	Holds the R11 limit on both legs on a bare concrete floor with no wall contact	Pull test on the anchored tail block	Met on paper on a sound slab: 0.45 of recommended anchor loads
R8	Setup time	Three people set up in 20 min or less	Timed trial	Not verifiable at TRL 3: about 17 min estimated
R9	Portability	Complete kit fits in a car boot; heaviest part 25 kg (55 lb) or less	Weigh and pack test	At risk: heaviest lift 24.7 kg; frame 890 x 663 x 814 mm needs an estate car or small pickup
R10	Cost	Prototype parts at or under the USD 2,000 value-engineering target (a hypothetical control target)	Costed bill of materials	Met on paper, within the target: USD 887.50 (USD 1,112.50 under)
R11	Overload limit	Rope tension limited to 3,000 N whatever force the crew applies	Shear pin release test on the capstan	Met on paper: shear pin releases at 2,451 N (1,961 to 2,941 N)
R12	Holds when let go	The drum holds the load in either direction when the cranks are released	Release test at working pull	Met by design: two ratchet wheels and pawls on the drum shaft

■ Requirements at risk or not met

- **R1 (not met).** Hand power sets the haul speed. Ways to close the gap belong to TRL 4 trials: a second box on the return leg so both strokes carry mud, or a crew of six with two at the cranks and four loading. Both change what the product is, so they are not made here.

- **R9 (at risk).** The frame is 25 mm inside the mass limit and too tall for some small hatchbacks. A bolted frame that packs flat is a later option.

■ Assumptions

- Most flood mud is fluid enough to be scraped and dragged, not dug; very stiff or deep deposits may still need shovels first.
- The building has a sound concrete slab at least 100 mm thick for the tail block anchors; where it does not, SiltHaul is not used in that room.
- Volunteers accept a short briefing on rope safety.
- A safe dump point exists outside within rope reach, and an anchor point (a sound tree or a parked vehicle) exists behind the capstan.